

ABSTRACT

The agriculture sector plays an important role in global economic stability and food security, particularly in countries such as India and China where paddy (rice) is not only a major primary food but also a major economic driving force. Crop diseases, especially in paddy fields, pose a significant risk to agricultural productivity and food security. Timely and accurate detection of these diseases is significant to prevent and mitigate crop losses and sustain rural economies. However, Traditional methods of disease detection primarily based on manual observation are labor intensive, inaccurate, and unsuitable for large-scale use. Also, traditional machine learning techniques often lack the ability to capture the complex visual patterns found in plant diseases, which makes them less effective in real-world agriculture.

To address these limitations, this project proposes a deep learning-based hybrid model that combines ResNet50 and Vision Transformer (ViT) architectures. ResNet50 model is able to learn fine-grained local spatial features using convolutional filters, and ViT model is able to learn long-range dependencies and global context relationships using self-attention mechanisms. By fusing the outputs of both networks together, the model captures a deeper and more comprehensive representation of disease features.

The system pipeline includes advanced preprocessing methods such as resizing, normalization, and contrast adjustment, followed by real-time data augmentation methods during training for improved generalization. These data augmentations such as flipping, rotation, and brightness adjustment help improve model's generalization ability by simulating diverse field conditions.

Extracted from ResNet50 and ViT features are concatenated and passed through fully connected layers, leading to a SoftMax classifier that accurately predicts the class of the disease with high accuracy. This hybrid architecture not only improves detection performance but also offers scalability and real-time applicability in precision agriculture.